

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, JavaScript

Tools/Frameworks: TensorFlow, PyTorch, FastAPI, Flask, Docker, Kubernetes, TensorRT, OpenAI API, SAP BTP, SAP Integration Suite, Airflow, Hive, PySpark

Competence: Machine Learning, Deep Learning, MLOps, Generative AI, Data Engineering, Data Analysis, Cloud Technologies, System Integration, APIs, Webhooks

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with PhD researchers to co-author a publication for the FLOMEKO conference.
- Developed dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x using advanced AI algorithms.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute costs and latency by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities, enhancing knowledge sharing and collaboration.
- Created POCs using YOLO, few-shot learning, and CNNs to demonstrate AI capabilities for client projects.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application providing AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and state-of-the-art LLM tools to enhance user interaction.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and ETL pipelines for structured and unstructured data, enabling efficient data analysis.
- Automated data ingestion and cleaning workflows, improving data quality and reducing manual effort.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Developed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Enhanced model accuracy by 4% through advanced feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Implemented a real-time Bitcoin price forecasting system using LSTM-based deep learning models and time-series analysis.
- Achieved high prediction accuracy, enabling informed decision-making for cryptocurrency trading strategies.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to create the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI-driven music creation.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in Machine Learning, Deep Learning, and AI-driven solutions for enterprise applications.
- Engaged in research on AI workflows, focusing on innovative applications of Generative AI and SAP integration.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence.
- Conducted a thesis on real-time Bitcoin price forecasting using LSTM models.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python